'''
LP-III Assignment 1 (Group A - DAA)
Write a program non-recursive and recursive program to calculate Fibonacci numbers and analyze their time and space complexity.
Name: Tanmay Sandip Bothara
Roll No: 23 Batch: BE_Comp_A1
'''

```
# Non-Recursive Fibonacci
count1 = 0
n = int(input("Enter No. Of Terms (For Non-Recursive): "))
count1 = count1 + 1
a = 0
count1 = count1 + 1
b = 1
count1 = count1 + 1
print("Non-Recursive Fibonacci Series: ")
count1 = count1 + 1
for i in range(n):
    count1 = count1 + 1
    print(a, end=" ")
    count1 = count1 + 1
    c = a + b
    count1 = count1 + 1
    a = b
    count1 = count1 + 1
    b = c
    count1 = count1 + 1
count1 = count1 + 1
print("\nNon-Recursive Count: ", count1)
```

'''
OUTPUT:-
Enter No. Of Terms (For Non-Recursive): 5
Non-Recursive Fibonacci Series:
0 1 1 2 3
Non-Recursive Count: 30
'''

```
# Recursive Fibonacci
count2 = 0
def fibonacci(n):
    global count2
    count2 = count2 + 1
    if n <= 1:
        count2 = count2 + 1
        return n
    count2 = count2 + 1
    return fibonacci(n - 1) + fibonacci(n - 2)
n = int(input("Enter No. Of Terms (For Recursive): "))
count2 = count2 + 1
print("Recursive Fibonacci Series:")
```

```
count2 = count2 + 1
for i in range(n):
    count2 = count2 + 1
    print(fibonacci(i), end=" ")
print("\nRecursive Count:", count2)
'''
OUTPUT:-
Enter No. Of Terms (For Recursive): 5
Recursive Fibonacci Series:
0 1 1 2 3
Recursive Count: 45
'''
```